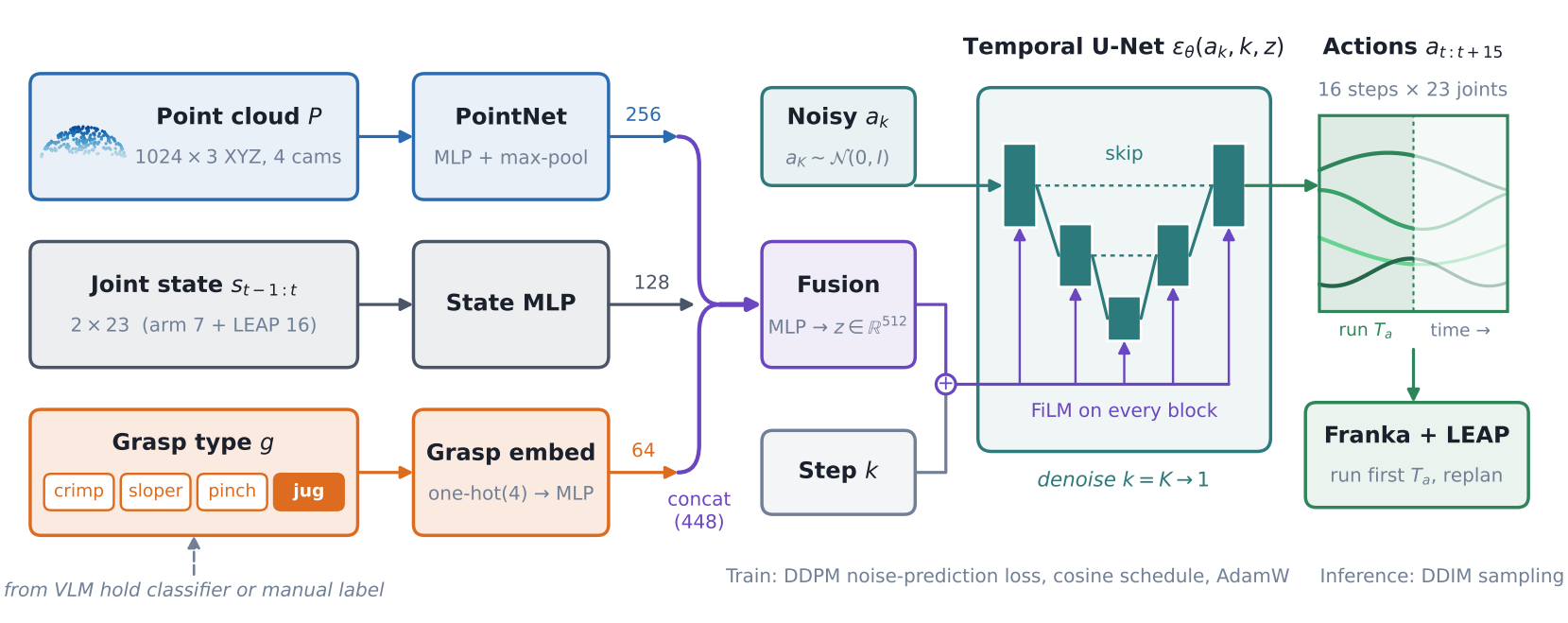

Point cloud P
1024 × 3 XYZ, 4 cams
PointNet
MLP + max-pool
256
Joint state $s_{t-1:t}$
2 × 23 (arm 7 + LEAP 16)
State MLP
128
Grasp type g
crimp
sloper
pinch
jug
Grasp embed
one-hot(4) → MLP
64
concat
(448)
from VLM hold classifier or manual label
Noisy a_k
$a_K \sim \mathcal{N}(0, I)$
Fusion
MLP → $z \in \mathbb{R}^{512}$
Step k
Temporal U-Net $\varepsilon_\theta(a_k, k, z)$
skip
FiLM on every block
denoise $k = K \to 1$
Actions $a_{t:t+15}$
16 steps × 23 joints
run T_a
time →
Franka + LEAP
run first T_a, replan
Train: DDPM noise-prediction loss, cosine schedule, AdamW
Inference: DDIM sampling